Database Schema — Gestion Stock Halal Distribution

This document describes the database tables, their columns, data types, constraints, and relationships for the **Halal Distribution Stock Management** application. Inferred from `src/data/mockData.js` and `src/services/api.js`.

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1. Enumerations (ENUMs)

`user_status`

Value	Description
`ACTIVE`	User is active
`INACTIVE`	User is deactivated

`production_status`

Value	Description
`AVAILABLE`	Lot is ready and available
`IN_PROGRESS`	Production is currently ongoing
`COMPLETED`	Production lot is completed

sale_status

Value	Description
EN_ATTENTE	Sale is pending
VALIDEE	Sale is validated
ANNULEE	Sale is cancelled

invoice_status

Value	Description
PAID	Invoice is paid
PENDING	Invoice is pending
UNPAID	Invoice is unpaid

movement_type

Value	Description
ENTREE	Stock incoming (production, return)
SORTIE	Stock outgoing (sale, loss)

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2. Table: users

Stores system user accounts with role-based access control.

Column	Type	Constraints	Description
id	INTEGER	PK, AUTO_INCREMENT	Unique identifier
nom	VARCHAR(100)	NOT NULL	Last name
prenom	VARCHAR(100)	NOT NULL	First name
email	VARCHAR(255)	NOT NULL, UNIQUE	Email address (used for login)
access_code	VARCHAR(20)	NOT NULL, UNIQUE	Access code (used for login)
role	ENUM	NOT NULL	ADMIN, SALES, STOCK
status	ENUM	NOT NULL, DEFAULT ACTIVE	user_status
date_creation	DATE	NOT NULL	Account creation date

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3. Table: `suppliers` (Fournisseurs)

Stores halal meat suppliers.

Column	Type	Constraints	Description
`id`	INTEGER	PK, AUTO_INCREMENT	Unique identifier
`nom`	VARCHAR(255)	NOT NULL	Supplier company name
`adresse`	TEXT	NULLABLE	Full postal address
`contact`	VARCHAR(50)	NULLABLE	Phone number
`email`	VARCHAR(255)	NULLABLE	Contact email address

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4. Table: `clients`

Stores customer/buyer accounts (butcher shops, restaurants, etc.).

Column	Type	Constraints	Description
`id`	INTEGER	PK, AUTO_INCREMENT	Unique identifier
`nom`	VARCHAR(255)	NOT NULL	Client business name
`adresse`	TEXT	NULLABLE	Full postal address
`contact`	VARCHAR(50)	NULLABLE	Phone number
`email`	VARCHAR(255)	NULLABLE	Contact email address
`type`	VARCHAR(100)	NULLABLE	Client type (e.g., `Boutique`, `Entreprise`, `Point de vente local`)

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5. Table: `articles`

The product catalog — all halal meat products sold by the company.

Column	Type	Constraints	Description
`id`	INTEGER	PK, AUTO_INCREMENT	Unique identifier
`reference`	VARCHAR(20)	NOT NULL, UNIQUE	Product reference code (e.g., `ART-001`)
`nom`	VARCHAR(255)	NOT NULL	Product name
`prix`	DECIMAL(10,2)	NOT NULL	Unit selling price (€)
`unite`	VARCHAR(20)	NOT NULL	Unit of measure (e.g., `kg`)

Column	Type	Constraints	Description
`categorie`	VARCHAR(100)	NOT NULL	Product category (e.g., `Volaille`, `Boeuf`, `Agneau`, `Charcuterie`)
`seuil_minimum`	INTEGER	NOT NULL, DEFAULT 0	Minimum stock alert threshold

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6. Table: `production_lots`

Tracks halal-certified production batches (lots) of articles.

Column	Type	Constraints	Description
`id`	INTEGER	PK, AUTO_INCREMENT	Unique identifier
`numero_lot`	VARCHAR(30)	NOT NULL, UNIQUE	Lot number (e.g., `LOT-2024-001`)
`article_id`	INTEGER	FK → `articles.id`	The article produced in this lot
`quantite`	DECIMAL(10,2)	NOT NULL	Quantity produced
`status`	ENUM	NOT NULL	`production_status`
`date_production`	DATE	NOT NULL	Date of production
`certificat_abattage`	VARCHAR(100)	NULLABLE	Halal slaughter certificate reference

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7. Table: `stock_movements` (Mouvements de Stock)

The **single source of truth** for all stock levels. Stock is computed dynamically as `SUM(ENTREE) - SUM(SORTIE)`.

Column	Type	Constraints	Description
`id`	INTEGER	PK, AUTO_INCREMENT	Unique identifier
`article_id`	INTEGER	FK → `articles.id`	The article affected by this movement
`type`	ENUM	NOT NULL	`ENTREE` (in) or `SORTIE` (out)
`quantite`	DECIMAL(10,2)	NOT NULL	Quantity moved
`motif`	VARCHAR(100)	NOT NULL	Reason: `Production`, `Vente`, `Perte`, `Annulation Vente`
`date`	DATE	NOT NULL	Date of the movement

Note: There is no separate `stock` table. Current stock = `Σ ENTREE - Σ SORTIE` from this table, filtered by `article_id`.

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8. Table: `sales` (Ventes)

Stores sales orders (commandes de vente) placed by clients.

Column	Type	Constraints	Description
`id`	INTEGER	PK, AUTO_INCREMENT	Unique identifier
`numero`	VARCHAR(30)	NOT NULL, UNIQUE	Sale number (e.g., `VNT-2024-001`)
`client_id`	INTEGER	FK → `clients.id`	The client who placed the order
`adresse_livraison`	TEXT	NULLABLE	Delivery address
`date_livraison`	DATE	NULLABLE	Expected delivery date
`total`	DECIMAL(10,2)	NOT NULL, DEFAULT 0	Total order amount (€)
`status`	ENUM	NOT NULL, DEFAULT `EN_ATTENTE`	`sale_status`
`date_creation`	DATE	NOT NULL	Date the sale was created

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9. Table: `sale_items` (Articles de Vente)

Line items for each sale order. A sale can contain multiple articles.

Column	Type	Constraints	Description
`id`	INTEGER	PK, AUTO_INCREMENT	Unique identifier
`sale_id`	INTEGER	FK → `sales.id`	The parent sale order
`article_id`	INTEGER	FK → `articles.id`	The article sold
`quantite`	DECIMAL(10,2)	NOT NULL	Quantity sold
`prix_unitaire`	DECIMAL(10,2)	NOT NULL	Unit price at time of sale
`total`	DECIMAL(10,2)	NOT NULL	Line total (`quantite * prix_unitaire`)

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10. Table: `invoices` (Factures)

Invoices automatically generated when a sale is validated (`VALIDEE`).

Column	Type	Constraints	Description
`id`	INTEGER	PK, AUTO_INCREMENT	Unique identifier
`numero`	VARCHAR(30)	NOT NULL, UNIQUE	Invoice number (e.g., `FAC-2024-001`)
`sale_id`	INTEGER	FK → `sales.id`	The linked sale order
`client_id`	INTEGER	FK → `clients.id`	The client to invoice
`montant_total`	DECIMAL(10,2)	NOT NULL	Total invoice amount (€)
`status`	ENUM	NOT NULL	`invoice_status`
`date_emission`	DATE	NOT NULL	Invoice issue date
`date_echeance`	DATE	NULLABLE	Payment due date

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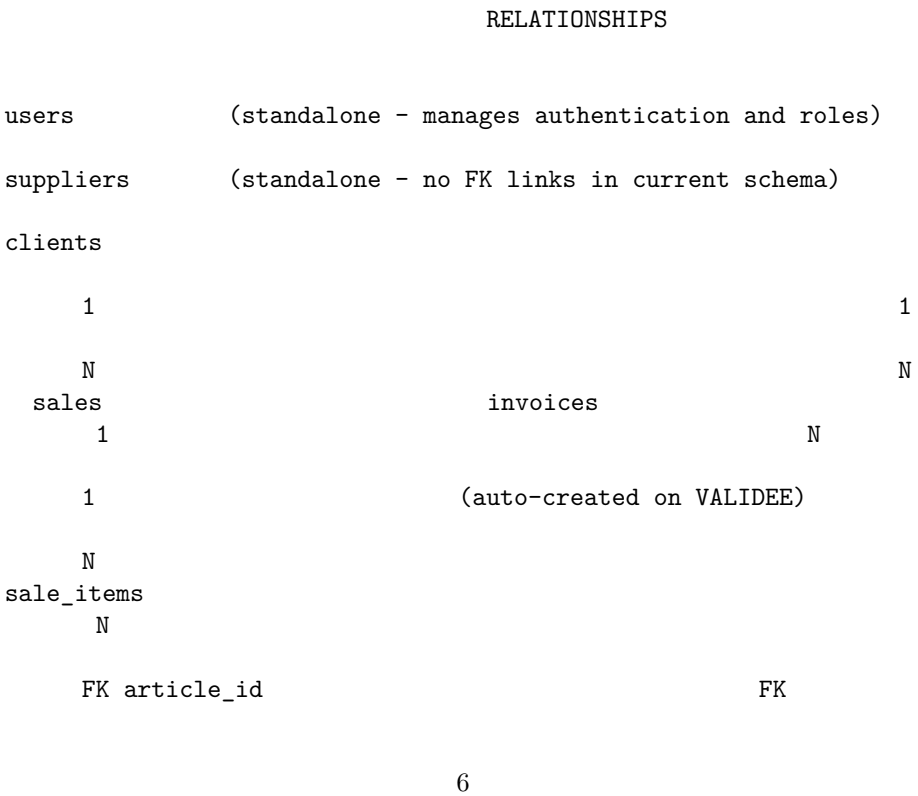
11. Relationships Diagram



```
articles
       1

                              production_lots (N)
       (article_id)

                              stock_movements (N)
        (article_id - single source of truth for stock levels)
```

Summary of Foreign Keys

Table	Column	References	Type
production_lots	article_id	articles.id	Many-to-One
stock_movements	article_id	articles.id	Many-to-One
sales	client_id	clients.id	Many-to-One
sale_items	sale_id	sales.id	Many-to-One
sale_items	article_id	articles.id	Many-to-One
invoices	sale_id	sales.id	One-to-One
invoices	client_id	clients.id	Many-to-One

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12. Business Rules

These rules are enforced at the application/service layer (`src/services/api.js`):

Rule	Description
Stock Calculation	Current stock level is never stored directly. It is always computed as `Σ ENTREE - Σ SORTIE` from `stock_movements` for each article.
Sale Validation	A sale can only be set to `VALIDEE` if all `sale_items` have sufficient stock available.
Auto Stock Deduction	When a sale is validated (`VALIDEE`), a `SORTIE` stock movement is automatically created for each line item.
Auto Invoice Creation	When a sale is validated (`VALIDEE`), an invoice (`UNPAID`) is automatically created and linked to that sale.
Sale Cancellation (Stock Restore)	When a validated sale is cancelled (`ANNULEE`), `ENTREE` stock movements (`Annulation Vente`) are created to restore the deducted stock.
Stock Alert	An alert is triggered for an article when `stock_actuel <= seuil_minimum`.

Rule	Description
Minimum Stock Guard	Creating a `SORTIE` movement is blocked if `quantite > stock_actuel` for that article.
Halal Certification	Each production lot must carry a `certificat_abattage` reference for traceability.
User Login	Login requires both a valid `email` AND `access_code`, and the user's `status` must be `ACTIVE`.